

Company Queries I

Time limit: 1.00 s

Memory limit: 512 MB

A company has n employees, who form a tree hierarchy where each employee has a boss, except for the general director.

Your task is to process q queries of the form: who is employee x 's boss k levels higher up in the hierarchy?

Input

The first input line has two integers n and q : the number of employees and queries. The employees are numbered $1, 2, \dots, n$, and employee 1 is the general director.

The next line has $n - 1$ integers $e_2, e_3, \dots, e_n$: for each employee $2, 3, \dots, n$ their boss.

Finally, there are q lines describing the queries. Each line has two integers x and k : who is employee x 's boss k levels higher up?

Output

Print the answer for each query. If such a boss does not exist, print -1 .

Constraints

- $1 \leq n, q \leq 2 \cdot 10^5$
- $1 \leq e_i \leq i - 1$
- $1 \leq x \leq n$
- $1 \leq k \leq n$

Example

Input	Output
5 3 1 1 3 3 4 1 4 2 4 3	3 1 -1